

Rave Generation

PRESENTS

FILTERBANK



USER MANUAL

1.0 Introduction

Welcome to Rave Generation: Filterbank. This state-of-the-art audio plugin brings your tracks to life with advanced analog filter modeling, sophisticated modulation systems, and rich distortion characteristics. Designed for producers, sound designers, and performers who demand exceptional sound quality and creative flexibility, Filterbank delivers warmth and character within a precision-engineered digital environment.

Filterbank is the result of extensive research into analog circuit behaviour, modelling the subtle non-linearities and component interactions that make analogue filters so coveted. Whether you're crafting experimental soundscapes, adding movement to static sounds, or bringing aggressive character to your mix, Filterbank provides the tools to elevate your productions with authentic analog warmth and character.

1.1 Key features

Rave Generation: Filterbank offers a comprehensive suite of features that empower you to shape sound with unprecedented detail and character:

- **Advanced analog modeling:** Features meticulously designed State Variable Filter architecture with non-linear feedback paths, capacitor simulation, and component interaction modeling for authentic analog sound.
- **Dual filter system:** Two independent filters with lowpass, bandpass, and highpass modes, each enhanced with ladder-style resonance feedback and precise blending capabilities. The filters can be arranged in parallel or serial configurations for maximum flexibility.
- **Multi-stage distortion:** Introduces genuine analog warmth or aggressive harmonic enhancement through multiple distortion algorithms including asymmetric processing, DC offset management, and dynamic component saturation. Includes four Neural modes powered by LSTM machine learning, each with its own saturation voicing.
- **Modulation matrix:** Integrates LFO, ADSR, and AR envelope generators with multiple trigger sources (audio, sidechain, MIDI) for evolving, dynamic control over all filter parameters.
- **Advanced envelope system:** Features Attack-Release (AR) amplitude control and Attack-Decay-Sustain-Release (ADSR) modulation with analog-inspired timing characteristics and multiple trigger modes including envelope follower functionality.
- **Thermal drift simulation:** Employs subtle modulation of harmonic distortion characteristics to recreate the "breathing" quality of analog hardware, adding organic variation to the sound over time.
- **Precision pitch tracking:** Dynamically adjusts filter frequencies based on input pitch detection, creating musical filter responses that complement the source material.
- **High-quality signal processing:** Employs 4x oversampling with sophisticated anti-aliasing for pristine audio quality, even with extreme settings.
- **MIDI integration:** Full MIDI CC compatibility with specialized note triggering system for live performance and sequenced modulation.
- **Real-time filter display:** A live frequency-response graph draws both filter curves and follows every filter control and the LFO in real time, with an on/off switch.
- **Fourteen LFO destinations:** The LFO is no longer tied to the filter cutoffs. It can be aimed at the cutoff of either filter or both, at the low/band/high blend, at the correction blend, at the resonance itself, at the parallel/serial routing, or at the dry/wet mix - turning a single modulator into a full matrix.

- **Global feedback loop:** Routes the filter output back into the input ahead of the drive stage, recreating the classic patch-cable feedback trick, with a dedicated tone control that damps the returning signal from dark to bright.
- **Pseudo-stereo width:** Spreads the two filters across the stereo field, from both filters on both sides through to a hard split of one filter per channel.

2.0 User interface

The interface of Rave Generation: Filterbank provides intuitive access to its powerful features. The layout follows the signal flow from input through filters to output, with modulation sources clearly organized for both creative experimentation and precise control.



2.1 Top row controls

INPUT: Controls the input gain and drive into the filter system. Higher settings introduce analog-style saturation and harmonic enhancement at the input stage, significantly affecting the overall character of the processing.

DRIVE ROUTE: The small icon above the INPUT label sets where the distortion sits in the signal chain. With the block on the LEFT of the line the signal is driven first and the filters then carve the resulting dirt - the classic arrangement, and how the plugin has always worked. With the block on the RIGHT the filters run first and their output drives the distortion, so the resonant peak is what distorts rather than being sculpted out of an already-dirty signal. The same DIST TYPE setting sounds substantially different either way; the effect is most obvious with resonance up. Note that FM takes its modulator from the driven signal, so FM is a little tamer in the second position.

FM (Frequency Modulation): Adjusts the amount of frequency modulation applied to both filters, allowing for complex timbral changes, sideband generation, and aggressive sonic transformations. The small icon next to the FM label is a clickable toggle that switches the modulation source between **Internal** (the plugin's input signal, post-overdrive) and **Sidechain** (an external signal routed via the sidechain input). The icon draws a signal path with a node on it: bare and dimmed when FM is fed internally, and with a branch joining the node from the side when the sidechain is driving it. By default FM uses the internal signal; click the icon to modulate from a sidechain source instead.

ADSR controls: The central ADSR section includes:

A (Attack): Controls the attack phase of the ADSR envelope, determining how quickly the modulation rises when triggered.

D (Decay): Sets the time for the ADSR envelope to fall from peak to sustain level after the attack phase completes.

S (Sustain): Determines the held level of the ADSR envelope while the trigger remains active.

R (Release): Controls how long the ADSR envelope takes to return to zero after the trigger ends.

LFO RATE: Controls the frequency of the LFO from extremely slow modulation through typical LFO rates to audio-rate modulation, enabling both subtle movement and FM-like effects.

LFO DEPTH: A bipolar control for LFO intensity. Center position provides no modulation. Turning right sweeps both filters in the same direction (in phase); turning left sweeps them in opposite directions (anti-phase), giving complementary modulation of the two filters. The ANTI switch (see 2.6) can lock this relationship regardless of which way the knob points.

LFO PHASE: Sets the LFO's start phase (0–360°). In LFO MODE = Restart, the LFO begins its cycle at the chosen point on each retrigger, letting you place the sweep where you want it; free-running, it offsets the phase globally. At 0° the behaviour is unchanged, so existing patches are unaffected.

LFO DEST: Selects what the LFO modulates - fourteen destinations in all. FILTER A+B is the classic behaviour (both cutoffs sweeping together), and FILTER A or FILTER B aims the sweep at one filter only. Beyond that: L-B-H A / B / A+B morphs the filter between lowpass, bandpass and highpass at a fixed frequency, giving a vocal, formant-like movement instead of a sweep; ORIG-CORR A / B / A+B animates the correction blend; RESO A / B / A+B pumps the resonance itself; PAR-SER slides the routing between parallel and serial; and DRY-WET modulates the mix. The ANTI switch (see 2.6) applies to every "B" destination, so Filter B can be made to move opposite Filter A on any of them. The default is FILTER A+B, so existing patches are unaffected.

SLEW: Rounds off the LFO's sharp edges. At zero the waveform is untouched; turning up softens the Square's flip and the Sawtooth's reset from an instant jump into a glide, and gently lags the Sine. The smoothing time is fixed (0.1–100 ms) rather than proportional to the rate, so fast LFOs are smoothed far more than slow ones - the same behaviour as a slew-limited analog circuit. Works on every waveform and every LFO destination.

AM (Amplitude modulation): Controls the depth of amplitude modulation, creating tremolo, ring-modulation, and gating effects depending on settings. The small icon next to the AM label works the same way as the one beside FM, switching the modulator source between **Internal** (Filter B's output, meaning RESO B strongly shapes the modulator character) and **Sidechain** (an external signal routed via the sidechain input).

PAR<>SER: Controls the routing of the dual filter system. Fully left (PAR) processes audio through both filters in parallel, while fully right (SER) routes the output of Filter A into Filter B for serial processing. Intermediate settings blend between these configurations.

DRY<>WET: Controls the balance between the bypassed (dry) and processed (wet) signals. At center position, both signals are mixed equally, while fully left bypasses all processing, and fully right applies 100% effect.

STEREO: Spreads the two filters across the stereo field. At zero both filters feed both sides, exactly as before. Turning up leans the left channel toward Filter A and the right toward Filter B, until at full they are hard-split - one filter per side. Because the two filters can be tuned to different frequencies, resonances and shapes, this turns a mono source into a wide, moving stereo image. It acts on the parallel signal path, so it has no effect with PAR<>SER set fully to SER, and it is ignored when the host provides a single output channel.

FEEDBACK: Feeds the filter output back into the input, ahead of the drive stage, so the returning signal saturates and gains harmonics on every pass. Low settings thicken and add body; higher settings build resonant ringing, howls and self-oscillation that can be played by the filter cutoffs. This is the classic patch-cable feedback trick, built in. The loop is soft-limited internally, so it stays controllable at every setting - but as with high resonance, use caution.

FB TONE: Shapes the signal inside the feedback loop, from dark (200 Hz) to bright (18 kHz). Dark settings give warm, rounded, dub-style feedback; bright settings let the highs keep circulating for screaming, metallic resonance. It only affects what is fed back, not the direct output.

2.2 Switch controls

BRIGHT/NORMAL/DARK: A three-position switch that controls pre-filter equalization. BRIGHT lifts the high frequencies before filtering for added air and bite, NORMAL maintains a neutral frequency response, and DARK rolls the highs away for a warmer, thicker character. The two sides are not symmetrical: BRIGHT is a gentle high shelf, while DARK is a full low-pass, so the cut side is the more drastic of the two.

ENV FLW/NORMAL/SOFT: A three-position switch controlling the trigger sensitivity for both AR and ADSR envelopes:

ENV FLW: Increases trigger sensitivity for better response to quiet sources

NORMAL: Standard trigger sensitivity for most audio material

SOFT: Reduces sensitivity to prevent false triggering with complex material

OCT SHIFT/NORMAL/PERF 5TH: A three-position switch for harmonic enhancement:

OCT SHIFT: Raises filter frequencies by one octave

NORMAL: No transposition

PERF 5TH: Raises both filters up a perfect fifth (x1.5)

PITCH/NORMAL/PITCH -OCT: Controls the pitch tracking function:

PITCH: Dynamically adjusts filter frequencies based on the detected pitch of the input

NORMAL: Disables pitch tracking

PITCH -OCT: Similar to PITCH but targets lower frequencies, ideal for bass material

TRACKING SOURCE: The small icon beside the pitch tracking switch chooses which signal the tracker listens to - Internal (the plugin's own input) or Sidechain (an external signal routed via the

sidechain input). On Sidechain a monophonic part such as a bassline can drive the filters while something else entirely is being filtered, giving cross-modulation between two separate sources. If nothing is patched into the sidechain the tracker falls back to the main input rather than tracking silence, and a mono sidechain drives both filter channels.

2.3 Filter controls

FILTER A: Sets the cutoff frequency for Filter A. This exponential control provides fine adjustment in the critical lower frequencies while allowing quick access to the full frequency spectrum.

RESO A: Adjusts the resonance feedback for Filter A. Higher settings emphasize frequencies near the cutoff point, creating peaks that can range from subtle enhancement to extreme self-oscillation.

SLOPE A: A two-position switch setting the steepness of Filter A. 12 dB/oct is the classic response - gentler and more open, letting more of the source through past the cutoff. 24 dB/oct runs the filter twice for a much tighter, more surgical cut, useful for isolating a narrow band or for sharper, more synth-like sweeps. Note that a steeper filter passes less overall, so 24 dB is quieter at low resonance settings; this is the slope itself, not a fault. The default is 12 dB, so existing patches are unaffected.

LBH 1 (Low-Band-High): A bipolar control that smoothly blends between lowpass (fully left), bandpass (center), and highpass (fully right) responses for Filter A.

BLH 1 (Band-Low-High Correction): A sophisticated correction control that modifies the filter response by adding or subtracting filter components. At center position (0), no correction is applied. Moving left (-B) reduces bandpass presence, while moving right (-B+LH) adds lowpass and highpass components to the bandpass.

RATIO: A 12-position rotary switch that defines the harmonic relationship between Filter A and Filter B. "FREE" allows independent control, while the numbered positions lock Filter B to specific musical intervals relative to Filter A.

MODE: A two-position switch above the RATIO knob. STEPPED uses 12 fixed positions. CONTINUOUS turns the knob into a smooth sweep of the harmonic ratio from 1x to 16x, for gliding, detuned movement between the two filters. In Continuous mode the FREE label reads CONT.

DIR: A two-position switch above the RATIO knob that sets the direction of the harmonic relationship. BELOW places Filter B below Filter A (the classic behavior). ABOVE places Filter B above Filter A, using the same intervals mirrored upward.

FILTER B: Sets the cutoff frequency for Filter B. When RATIO is active, this control is subordinate to FILTER A according to the selected harmonic relationship.

RANGE: A two-position switch above the FILTER B knob, active only when RATIO is engaged. OFF is the classic behavior (FILTER B has no function while synced). TRANSPOSE repurposes the FILTER B knob to slide the whole synced filter pair upward (up to two octaves) while keeping the harmonic interval between the filters intact.

RESO B: Controls the resonance for Filter B with a slightly different character than RESO A, allowing complementary tonal shaping when both filters are used together.

SLOPE B: Sets the steepness of Filter B, 12 or 24 dB/oct, exactly as SLOPE A. The two switches are independent, so one filter can stay wide and open while the other is tight and focused.

LBH 2: Provides the same filter type control as LBH 1 but for Filter B, enabling complex dual-filter configurations.

BLH 2: Provides the same correction functionality as BLH 1 but for Filter B.

FILTER DISPLAY: The graph in the centre of the lower panel draws the live frequency response of both filters (FILTER A blue, FILTER B teal). It follows every filter control in real time - cutoff, resonance, the L/B/H blend, the ORIG/CORR shaping, the RATIO link (with Direction and Range), the +Oct/+Quint transpose, the BRIGHT/DARK switch and the 12/24 dB SLOPE switches — and it animates in time with the LFO, following whichever destination LFO DEST is set to. It shows the set filter shape, not an audio spectrum. The GRAPH switch turns the display off or on.

TIPS: Shows or hides the hover hints. With TIPS on, holding the mouse over any knob or switch shows a one-line description of what it does in the hint bar; turn it off for a cleaner panel once you know your way around. Display only - it does not affect the sound.

2.4 Envelope and trigger controls

TRIG AR: Selects the trigger source for the AR envelope:

OFF: No external triggering (manual control only)

SIDECHAIN: Uses audio from the sidechain input for triggering

MIDI: Allows MIDI notes to trigger the AR envelope

A (Attack): Controls the attack time for the AR amplitude envelope, shaping how quickly the processed signal reaches full volume.

R (Release): Sets the release time for the AR envelope, determining how the processed signal fades out when the trigger ends.

HOLD: Sets how long an envelope stays open once triggered, from 1 ms up to 500 ms, before the release begins. Short settings give the snappiest response; longer settings keep the filter open through the body of a drum hit or a plucked note, and stop fast percussive material from retriggering the envelope into a chatter. Affects both the AR and the ADSR envelopes. At the minimum it matches the previous fixed 1 ms hold, so existing patches are unaffected.

2.5 Bottom panel controls

DIST TYPE: Choose between different distortion modes:

CLEAN: Minimal distortion with subtle enhancement

WARM: Analog-inspired saturation with even harmonic emphasis

CRUNCH: More aggressive distortion with complex harmonic content

EDGE: Maximum distortion with pronounced frequency-dependent character

DYNAMIC: Input-responsive distortion with signal-dependent bias and saturation memory, creating tube-like sag and harmonic interaction that evolves with playing intensity

NEURAL: Filterbank's own drive character, reproduced by an LSTM neural network. Dense, harmonically rich saturation across the gain range. **RAT:** Aggressive fuzz voicing in the tradition of the hard-clipping distortion pedals of the late 1970s. Fuzzy and mid-forward at high gain. **DS-1:** Tight crunch voicing in the tradition of the Japanese distortion pedals of the early 1980s. Hard edged and focused at high gain. **TS9:** Smooth, mid-pushed overdrive voicing in the tradition of the Japanese tube-style overdrive pedals of the early 1980s. Soft clipping with a strong midrange lift.

Note: All four neural modes bypass the algorithmic distortion entirely and run LSTM inference instead. Oversampling is disabled in these modes, as the networks already produce the full harmonic spectrum. Neural modes run hotter than the algorithmic ones, so use the **OUTPUT** knob to compensate.

TRIG ADSR: Selects the trigger source for the ADSR envelope.

OFF: No external triggering (manual control only)

SIDECHAIN: Uses audio from the sidechain input for triggering

MIDI: Allows MIDI notes to trigger the ADSR envelope

TRIG AR: Selects the trigger source for the AR envelope.

OFF: No external triggering (manual control only)

SIDECHAIN: Uses audio from the sidechain input for triggering

MIDI: Allows MIDI notes to trigger the AR envelope

ADSR SENS: Controls the audio threshold level (in dB) for ADSR triggering. Adjust this to fine-tune how the ADSR responds to your audio material.

ENV DEST: Selects what the ADSR envelope modulates - fourteen destinations, the same set the LFO has and in the same order. **FILTER A+B** is the classic behaviour, the envelope opening both filter cutoffs, and is the default so existing patches are unaffected. Beyond that the envelope can morph the L-B-H blend, so each hit sweeps the filter from lowpass through bandpass to highpass at a fixed frequency rather than sweeping its frequency; drive the **ORIG-CORR** correction blend; pump the resonance itself; slide the parallel/serial routing; or move the dry/wet mix. Depth comes from the existing **ADSR AMOUNT A** and **B** knobs rather than a control of its own - **AMOUNT A** drives the **A** destinations, **AMOUNT B** the **B** ones, and **AMOUNT A** drives the two shared destinations (**Par/Ser** and **Dry/Wet**) since neither belongs to a single filter. Those knobs are bipolar, so a negative amount closes on peaks instead of opening, and setting one positive and the other negative moves the two filters in opposite directions from the same hit. The LFO and the envelope may be aimed at the same destination, in which case their movements simply add.

AR SENS: Controls the audio threshold level (in dB) at which the AR envelope will trigger when using audio input as a trigger source. Lower values make the AR more sensitive to quiet sounds.

Note on the Sidechain input: The sidechain input is shared across the **FM** source toggle, **AM** source toggle, and the **AR/ADSR** sidechain triggers. A single sidechain signal can simultaneously

fire your envelopes and drive FM/AM modulation - useful for things like a kick triggering ADSR while also shaping the carrier with the kick's transient.

THD DRIFT: Controls the amount of thermal drift simulation, affecting how the distortion characteristics subtly change over time. Higher settings create more pronounced "breathing" effects typical of analog hardware.

NOISE: Introduces calibrated pink noise into the signal path, useful for adding texture, creating ambient atmospheres, or enhancing percussive elements.

LFO SYNC: Synchronizes the LFO frequency to your DAW's tempo with various rhythmic divisions (1/1, 1/2, 1/4, etc.), ensuring perfectly timed filter modulation.

OUTPUT: Adjusts the final output gain after all processing. This compensates for level changes introduced by filtering and distortion.

2.6 Additional controls

LFO MODE: A four-position switch selecting the LFO waveform and retrigger behaviour. RESTART restarts the LFO with each new trigger; NORMAL is a free-running sine; SAWTOOTH is a falling ramp with a hard reset, for rhythmic, pulsing sweeps; and SQUARE jumps instantly between two points, for stepped, gated, trill-like modulation. Both the sawtooth and the square are band-limited, so they stay clean and free of aliasing even at audio rates - and the SLEW control can soften their edges as much as you like.

ANTI: A three-position switch setting how the two filters move relative to each other under the LFO. NORMAL (default) keeps the classic behaviour — LFO DEPTH decides, so a right (positive) depth sweeps both filters together and a left (negative) depth sweeps them in opposite directions (anti-phase). IN-PHASE forces the filters to always sweep together, whichever way DEPTH points. ANTI-PHASE forces them to always sweep in opposite directions. Without this switch anti-phase is only reachable through negative depth; it lets you lock either relationship at any depth.

ADSR MODE: A three-position switch selecting between:

ADSR: Standard envelope generator mode with full ADSR control

FREEZE: Sustains the current envelope state regardless of trigger input

ENVELOPE FOLLOWER: Dynamically tracks the amplitude of the input signal

OVERSAMPLE: Toggles between standard processing and 4x oversampling. When enabled, this significantly reduces aliasing artifacts, particularly with high resonance settings and upper-frequency content, at the cost of increased CPU usage. Note: Oversampling is automatically disabled when any neural distortion mode is active (Neural, Rat, DS-1, TS9), as the networks already produce the full harmonic spectrum.

2.7 MIDI controls

When the AR Trigger or ADSR Trigger is set to MIDI mode, you can control the triggering behavior via MIDI note messages. These controls allow precise timing and triggering from your MIDI controller, integrating Filterbank into live performance or MIDI-based production setups.

MIDI Note	Function	Behavior
60 (C4)	Unblock ADSR	Allows ADSR to trigger from audio or MIDI.
61 (C#4)	Block ADSR	Prevents ADSR from triggering (audio or MIDI blocked).
62 (D4)	Unblock AR	Allows AR to trigger from audio or MIDI.
63 (D#4)	Block AR	Prevents AR from triggering (audio or MIDI blocked).
66 (F#4)	Normal Trigger ADSR	Triggers the ADSR envelope normally.
70 (A#4)	Normal Trigger AR	Triggers the AR envelope normally.
68 (G#4)	Normal Trigger Both	Simultaneously triggers both AR and ADSR envelopes.
65 (F4)	Trigger ADSR	Triggers ADSR with instant attack.
67 (G4)	Gate-Off ADSR	Forces gate-off of ADSR with instant release.
69 (A4)	Trigger AR	Triggers AR with instant attack.
71 (B4)	Gate-Off AR	Forces gate-off of AR with instant release.

Note on Octave Labeling:

- MIDI note 60 corresponds to Middle C (C4).
- Some DAWs (e.g., Ableton, Logic, Studio One) may label this note as C3. This is simply a labeling difference, and the functionality remains the same.

3.0 Technical overview

Filterbank employs several advanced analog modeling techniques to achieve its distinctive sound quality. This section provides insight into the technical innovations that make this plugin unique.

3.1 SVF implementation

At the heart of Filterbank is a State Variable Filter (SVF) architecture that accurately reproduces the behavior of analog hardware filters. Unlike typical digital filters, our implementation:

- Uses separate state variables that correspond to physical elements in analog circuits
- Provides simultaneous lowpass, bandpass, and highpass outputs with precise blending
- Employs bilinear transform prewarping for accurate frequency response at all settings
- Calculates coefficients dynamically based on cutoff frequency and resonance parameters

3.2 Analog circuit modeling

Several techniques are used to faithfully emulate analog circuit behavior:

Diode pair simulation: The feedback path incorporates non-linear diode modeling that creates the characteristic asymmetric distortion found in analog filters, enhancing harmonic content and providing warmth.

Ladder resonance feedback: A recursive resonance system with the cascaded-stage behaviour of a transistor ladder topology, giving the distinctive analogue resonance character when pushed to self-oscillation.

Non-linearity control: A specialized non-linearity factor parameter allows smooth transition between clean digital response and analog-like behavior, particularly important at high resonance settings.

Resonance delay modeling: Filter parameters incorporate subtle delays that simulate component-level interactions, creating a more organic response to rapid parameter changes.

3.3 Capacitor simulation

The distinctive response of analog filters is partly due to capacitor charge/discharge characteristics:

- Different time constants for charging and discharging phases create non-linear response curves
- Switched capacitor behavior emulates how analog circuits naturally shift between states
- Sign-dependent reset behavior accurately captures real capacitor response to signal polarity inversions
- Dynamic parameter adjustment based on resonance and frequency settings creates the characteristic "squeeze" heard in analog filters

3.4 Distortion and saturation

Filterbank incorporates multiple stages of analog-inspired distortion:

- Signal path saturation based on the hyperbolic tangent curve creates natural compression effects
- Asymmetric distortion processing generates the even harmonics characteristic of analog hardware
- DC offset management enhances harmonic character while preventing unwanted artifacts
- Multiple distortion modes (Clean, Warm, Crunch, Edge, Dynamic, Neural, Rat, DS-1, and TS9) provide a wide range of tonal options, from algorithmic saturation to neural network voicings

3.5 Thermal drift and noise

To capture the subtle imperfections that give analog equipment its character:

- THD (Total Harmonic Distortion) drift simulation uses low-frequency modulation to create the "breathing" quality of analog circuits
- Multi-stage filtered noise generation produces authentic pink noise for added texture
- Adaptive noise that responds dynamically to signal levels emulates how real components behave
- Weight-based distortion coloration creates distinctive sonic signatures at different frequency ranges

3.6 Advanced modulation system

The modulation system builds upon the analog modeling techniques:

- LFO modulation with four waveforms (Sine, Sawtooth, Square and a retriggered Sine), band-limited so they stay clean at audio rates, with a slew control for softening the edges and fourteen selectable destinations
- ADSR envelope with analog-inspired attack, decay, sustain, and release characteristics
- Envelope follower mode that dynamically tracks input signal for organic filter modulation
- Sophisticated cross-modulation between filters for complex, evolving sounds
- Pitch tracking that adapts filter frequency based on input signal's harmonic content

3.7 Analog-inspired gain staging

Proper gain staging is crucial to analog sound quality:

- Input gain structure dynamically affects distortion characteristics and filter behavior
- Resonance compensation prevents excessive gain at high resonance settings
- Adaptive gain scaling based on filter settings maintains consistent output levels
- Analog-inspired dynamic range compression at signal path transitions
- Subtle high-frequency enhancement through carefully calibrated high-shelf filtering

3.8 Component interaction modeling

Real analog circuits exhibit complex interactions between components:

- Cross-modulation between state variables creates subtle harmonic interactions
- Deliberate asymmetry in feedback paths mirrors real circuit behavior
- Accurate modeling of component saturation under varying signal conditions
- Dynamic interaction between resonance and cutoff frequency settings
- Self-oscillation characteristics that behave like genuine analog counterparts

Note on self-oscillation: Use caution when using high resonance.

4.0 Installation & troubleshooting

4.1 System requirements

Before installing Rave Generation: Filterbank, please ensure that your system meets the following requirements:

- Operating system:
 - macOS 10.13 or later
 - Windows 10 or later
- Software: Digital Audio Workstation (DAW) that supports VST3, or AU plugins (e.g., Ableton Live, Logic Pro, Studio One, FL Studio, etc.).
- Processor: Intel Core i5 (or equivalent) or higher for optimal performance.
- RAM: 4 GB minimum (8 GB or more recommended for larger projects).
- Disk Space: 200 MB of free disk space for installation.

4.2 Installation process

1. Download the installation file from the official website or the platform where you purchased the plugin.
2. Run the installer and follow the on-screen instructions.
3. Launch your DAW and locate Rave Generation: Filterbank in your plugin list.
4. If prompted, activate the plugin using the license key provided upon purchase.

4.3 Troubleshooting

If you encounter any issues during installation or operation, try the following solutions:

- Plugin not showing in DAW: Ensure that the plugin folder path is correctly set within your DAW's plugin manager.
- Activation issues: Double-check your internet connection and ensure you are entering the correct license key.

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